

No evidence for IKEA effect in children's long-term interactions with social robots

Abstract

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1 Introduction and related work

Social robots have shown significant potential to support cognitive and affective learning across diverse educational contexts (Belpaeme et al., 2018). While typically cast as teachers, tutors, or peers, their roles continue to expand to for example socio-emotional support or conversational roles (Rohlfing et al., 2022). However, widespread school deployment is hindered by factors including financial burden, technical complexity, user acceptance, and long-term engagement (Leisten, Graw, et al., 2026; Leite et al., 2013; Rosén et al., 2022; Woo et al., 2021). One possible solution to these challenges is the use of inexpensive, simple “do-it-yourself” (DIY) social robotic toolkits. Recent platforms such as the Reachy mini (Pollen Robotics & Hugging Face, 2026) and the Blossom robot (Suguitan & Hoffman, 2019) offer social behaviors (e.g., movement, speech, facial expressions) at a fraction of the cost (\$100-\$500 USD) of commonly used robots like NAO or Pepper (\$8,000-\$20,000 USD; Aldebaran Robotics, 2023; Mahdi et al., 2022). These toolkits often require users to engage in the assembly process, including 3D printing, wiring, motor calibration, and programming. This transition from mere consumers to creators of AI technology may foster students' digital literacy and increase long-term engagement, but it may also reduce the socialness perception of these agents and thereby reduce their educational effectiveness (Goldman & Poulin-Dubois, 2024). To explore the potential of DIY robots to support children's learning, it is crucial to understand the possible trade-off between the psychological benefits of authorship and the potential loss of socialness attribution.

The process of assembly is well-documented to improve users' evaluation of objects. In 2012, Norton and colleagues (2012) identified the “*IKEA effect*”, where manual assembly of purchased IKEA furniture increased consumer's satisfaction of it. The IKEA effect has also been confirmed in a recent metaanalysis (Pelled et al., 2026). Likewise, the “*I designed it Myself*” effect suggests that the feeling of authorship enhances perceived value through a sense of ownership and accomplishment (Franke et al., 2010; Sun & Sundar, 2016). In Human-Robot interaction (HRI), numerous studies demonstrated that the sense of ownership and feelings of accomplishment derived from assembling a robot impact the attachment to it and how people perceive it (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016). For example, Groom and colleagues (2009) show that adults reported lower maliciousness, higher friendliness, and stronger team feelings with a robot they assembled versus a robot assembled by others. In Sun and Sundar's work (2016), adults evaluated both the self-assembled robot and the interaction with it as more positive than a preassembled robot. Robot-assembly has also been linked to increased self-efficacy, personal engagement, emotional attachment, and empathy (Groom et al., 2009; Huang et al., 2013; Tsumura & Yamada, 2025), leading to a reluctance to dismantle the robot (Huang et al., 2013). These findings highlight the positive impact of robot-assembly on users' attachment, which in turn may foster increased intention to use that robot, as well as the long-term engagement needed for successful robot deployment (Zhang & Yu, 2025).

Besides the benefits of psychological ownership, assembly may also affect how a user perceives and rationalizes a robot's behavior. According to Dennett's (1971) theory of intentionality, humans adopt different mental stances to predict a system's behavior: (i) the physical stance, where people predict behavior based on physical states and natural laws, (ii) the design stance, where people predict behavior based on the system's intended functional design and their knowledge of it, and (iii) the intentional stance, where people treat the system as a social agent with intentional beliefs and desires. Both the physical and design stance require a clear understanding of a system's technical design or functionality (Dennett, 1971). However, current social robots are often designed as “black boxes” without clear indications about their functionality. Paired with children's lack of technical knowledge and their readiness to ascribe socialness to inanimate objects, this may promote the adoption of an

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intentional stance towards social robots (Beran et al., 2011; De Jong et al., 2020; Leisten et al., 2024; Leisten, Moffat, et al., 2026; Marchesi et al., 2019; UNESCO, 2021).

Adopting an intentional stance towards social robots may facilitate social attunement and robots' integration into everyday life (Perez-Osorio & Wykowska, 2020). For example, Wiese and colleagues (2012) found that people's social attention is affected by whether or not they take an intentional stance towards robots. Similar findings come from Hechler and colleagues (2025), who showed that believing in human-control of artificial systems improved adults' subjective experience in a collaborative task. However, adopting an intentional stance towards technology may not always be beneficial, as it could foster unhealthy emotional attachment or over-trust (Perez-Osorio & Wykowska, 2020), and even results in higher (bodily) wariness in interactions with robots (Broadbent et al., 2011). These risks are particularly problematic for children, who often lack the necessary AI literacy to interact safely with AI technology (Leisten, Moffat, et al., 2026; Mertala & Fagerlund, 2024; UNESCO, 2021).

The adoption of an intentional stance, and with it the attribution of socialness, is a dynamic, multidimensional process rather than a binary choice (Cross et al., 2016; Hortensius & Cross, 2018; Leisten et al., 2025; Liberman-Pincu et al., 2023). Because social perception is composed of multiple dimensions (e.g., described as experience and agency, warmth and competence, human-nature and human-uniqueness, and other multidimensional approaches; Cross & Ramsey, 2021; Fiske et al., 2007; Gray et al., 2007; Henschel et al., 2020; Leisten et al., 2025), a robot may be perceived as intentional or social on some levels while being understood as a functional tool on others. Consequently, robot-building may support the adoption of a design stance by learning about the system's underlying functions (Hitron et al., 2019; C. L. van Straten et al., 2020), while the time and effort spent on assembly may create feelings of ownership and attachment (Sun & Sundar, 2016). This unique combination of social processes may make social DIY robots particularly well-suited to support responsible and beneficial school deployment.

Despite the potential tension between psychological ownership and socialness perception, very few studies have investigated the effect of assembly on children's perceptions of robots. Williams and colleagues (2019) collected pre-schoolers' (4-6 years) perceptions of a robot's theory of mind, agency, intelligence, and socialness before (pre-test) and after (post-test) building a PopBots robot (consisting of lego blocks, a phone, and a tablet) and participating in AI learning activities. From pre- to post-test, children's perception of a robot's intelligence and socialness increased. These effects were moderated by children's AI knowledge: children with higher AI knowledge were more likely to perceive robots as intelligent and social. Su and Yang (2024) extended these results by assessing how an AI literacy curriculum may affect pre-schoolers' perceptions of robots using the same measures as Williams and colleagues (2019). The authors found that children perceived a robot as more intelligent and social after learning about AI. This research sets the stage for investigations into the influence of robot-building on older children and adolescents' perceptions of robots' socialness. Adolescents are an especially relevant stakeholder group in this context, as they are at particular risk from unsafe AI interactions, due to their high usage of the technology but insufficient AI education and risk awareness (Sarikakis & Chatziefrimidou, 2025; UNESCO, 2021). To evaluate the potential of robot-assembly kits to overcome robot deployment challenges (i.e., costs, technical complexity, long-term engagement) it is important to assess the influence of robot-assembly on middle-school children's and adolescents' perceptions, attachment, and intention to use this robot.

1.1 The present study

In this study, we sought to investigate how building a social robot across multiple classroom lessons influences older children's and adolescents' perceptions of, attachment to, and intention to use a social robot. In one of the first long-term educational HRI study with adolescents (Matheus et al., 2025), we collected survey data at three time points during a 7-weeks educational curriculum for older children and adolescents aged 12-16 years. To isolate the influence of the robot-building experience, we employed an active control group design with a *build group*, who built a robot, and a *control group*, who interacted with an already assembled robot. We assessed adolescents' perceptions of the robot they built (build group) or interacted with (control group), as well as their perceptions of two other robots they did not build/interact with. Additionally, we assessed adolescents' attachment and intention to use the robot they built/interacted with. We aimed to address the following research questions:

- **RQ1:** What is the effect of robot-building on adolescents' social perception of robots and does this differ between the robot they built/interacted with and other robots?
- **RQ2:** What is the effect of robot-building on adolescents' attachment and intention to use the robot?

In line with the above-reviewed work, we expected children in the build group to perceive the robot as more competent (**H1**), warmer (**H2**), safer (**H3**), and less discomforting (**H4**), and more anthropomorphic (**H5**), over time than in the control group. We also expected children in the build group to feel more attached to (**H6**) and willing to use (**H7**) the robot over time than children in the control group. We additionally preregistered exploratory analyses of interactions with adolescents' prior digital literacy knowledge, attachment to the robot, and the generalization of these effects to other robots.

2 Methods and materials

2.1 Participants

We recruited a convenience sample of two participating classes from a rural German comprehensive school, resulting in a sample of 61 children between 12-16 years (15 females, 42 males, 4 other/NA, $M_{age} = 13.40$, $SD = 0.75$). Participants were native German speakers. Informed parental consent was collected and children above 14 years provided additional written consent. The ETH Zurich's ethical committee approved the study (25 ETHICS-247).

2.2 Design and procedure

We employed a longitudinal, between-subjects, active control design conducted over a seven-week educational curriculum during children's regular Natural Sciences and Informatics class. The intervention was split into two phases: In phase 1 (weeks 1-4), children in the build group ($n = 31$) built their own social robot, *Marty* (Robotical, 2025), in randomly assigned teams of 3-4. Children stayed within their assigned teams throughout the entire curriculum. Children in the control group ($n = 30$) interacted with pre-assembled *Marty* robots, also in teams of 3-4. In phase 2 (weeks 5-7), both groups used their respective robots to engage in identical learning interactions, described in more detail below. We selected robot interaction as the control condition, to ensure that the interaction time was consistent between both groups. This allowed us to compare robot-assembly to typical robot deployment, namely to pre-assembled tools that students can readily interact with.

The seven-week curriculum covered social robotics, Artificial Intelligence (AI), Large-Language models (LLMs), and programming. Each weekly session included theoretical content (~20-50 minutes) and practical activities (~15-100 minutes) and lasted 140 minutes. The total experimental time in the classroom was 16 hours and 33 minutes. Participants were divided into two parallel tracks led by two separate teachers, due to the school's classroom set-up with two adjacent rooms. We first conducted two experimental groups in parallel (Nov – Dec 2025), followed by two parallel control groups (Jan – Mar 2026).

Children completed pre-, mid-, and post-test surveys (~15 minutes) in Week 1, Week 4, and Week 7 via Qualtrics (2024) surveys on iPads. In the surveys, children first watched 4-5-second-long videos of *Marty*, *NAO*, and *Blossom*, and rated each robot on seven social attributes using the robotic social attributes scale (RoSAS-SF; Carpinella et al., 2017; Neuenswander et al., 2025). We selected *NAO* and *Blossom* in addition to *Marty*, to provide examples of a prominently used, complex robot (*NAO*) as well as a simpler DIY robot (*Blossom*). The video stimuli were obtained from Youtube (Youtube, 2024) and muted to avoid confounding effects. The order of robots and attributes was counterbalanced. Next, children were presented with a picture of *Marty* and asked to answer the specific object anthropomorphism scale (SOAS; David et al., 2025), the product-consumer attachment scale (Schifferstein & Zwartkruis-Pelgrim, 2008), and the intentional acceptance scale (De Jong et al., 2020). Scale and item order were counterbalanced across children. Lastly, children were asked about their subjective AI literacy knowledge and their objective digital literacy and additionally completed short feedback surveys (~10 minutes) at the end of each session, which we report elsewhere (XXX). Children

were naïve about the study objectives and were debriefed after the completion of the control group. Teachers received a 100€ voucher towards classroom materials for their classes' participation.

2.3 Marty robot

Marty is a medium-sized (~25 cm), humanoid robot, designed to support children's STEAM (Science, Technology, Engineering, Arts, Mathematics) education at a price-point of ~449€. Equipped with two LED eyes, eleven motors (eyes, arms, hips, knees, thighs), and five sensors (acceleration, tilt, position, colour, infrared), Marty's behavioural repertoire includes walking, turning, dancing, celebrating, speaking, playing music, eye-wiggling and ball-kicking. Marty features screen-free, block-based, or text-based coding (Python or C++ with Arduino), using the App or web-based version.

2.4 Curriculum

We developed a seven-week digital literacy curriculum centered around technical, AI, and programming literacy based on existing curricula and frameworks. The curriculum was designed to meet the following requirements: (i) be age and skill appropriate for 12-16-year olds, (ii) cover a relevant breadth of topics to serve as a pilot for future research while also providing practical and useful knowledge to children, and (iii) fit the logistical requirements of the participating teachers and the selected robot. While developing the curriculum, we constrained the amount of content per lesson, included practice and discussions, and set aside time for recall and repetition. The developed curriculum is available in English and German on GitHub ([link](#)) and would benefit from further improvement through iteration and future research. An overview of the 7-week curriculum can be seen in **Figure 1**.

We started the curriculum development from OECD's AI literacy framework (2025) and selected the level "*engaging with AI*" as our core focus, which requires learners to recognize AI's presence, evaluate its output's accuracy and relevance, and develop a fundamental understanding of technical foundations of AI. We selected this level as a first introduction to the topic, as children in our sample did not have previous AI education. To create the learning materials, we searched for existing open-source AI literacy curricula, and reviewed 15 curricula in more detail (**Table 1**). From these, we identified the topics covered, and selected those that matched with the OECD framework and the needs of the present study, namely: (i) what is AI, (ii) what is Machine Learning (ML), (iii) what are LLMs and how do they work, (iv) biases and hallucinations, (v) how to use LLMs and prompt engineering, (vi) AI ethics, and (vii) AI sustainability. We also added elements that we considered important and that were not part of any other curriculum. Specifically, we integrated social robots, sustainability of AI, deception by LLMs, and LLM safeguards. Having identified the curriculum topics, we refined and adapted the curriculum in an iterative process with the participating teachers to their needs (e.g., programming, sensors, microcontrollers, C++), and existing teaching materials.

Table 1. Reviewed curricula descriptions

Curriculum	Source	Content	Age group
Computingeducation TINAs	(Computingeducation, 2025)		10-19 years
Lernreise	(Day of AI, 2026)		12-18 years
Day of AI – Brain behind the Bot	(MIT RAISE, 2026)		11-14 years
MIT DAILY	(Common sense media, 2026)		11-18 years
Common Sense Media	(aiEDU, 2026)		11-18+ years
aiEDU	(AI4GA, 2026)		14-18 years
Georgia AI4GA	(Roteco, 2026)		14-18 years
ENARIS EPFL	(MIT Media Lab, 2026)		10-18 years
MIT AI Ethics	(AI4K12, 2026)		5-18 years
AI4K12	(ADAPT center, 2026)		15-16 years
AI Adapt / AI in my life	(Duke Center for Teaching and Learning, 2026)		18+
Duke learning innovation			

Harvard AI pedagogy	(AI Pedagogy Project, 2026)	18+
Toddle AI 101	(Toddle, 2026)	5-18 years

Regarding technical literacy and programming, we included an introduction to Marty’s components (i.e., motors, sensors, onboard computers, and wiring), and also introduced microcontrollers (i.e., Arduino) based on the teachers’ existing materials. We additionally included an introduction to programming, covering commands, parameters, variables, loops, if-else commands, and text-based programming in C++, again based on teachers’ already existing materials.

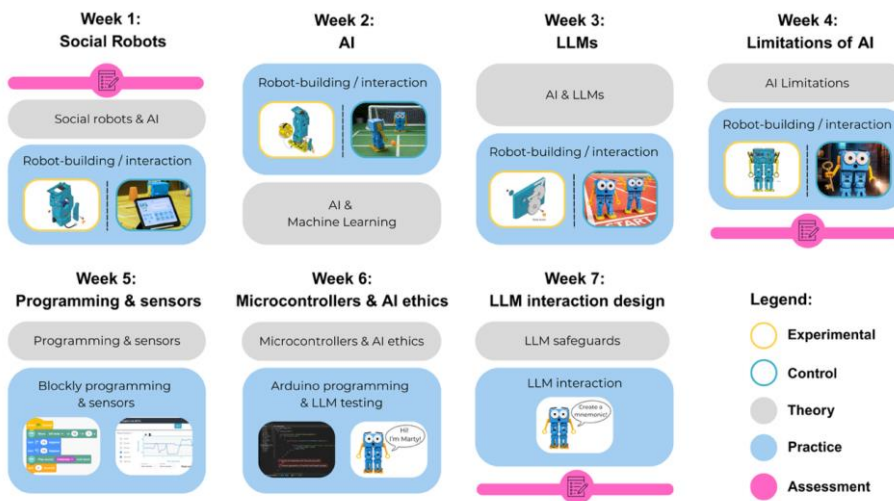


Figure 1. Schematic overview of the 7-weeks curriculum.

2.3.1 Week 1: Social robots

Both groups. Children were introduced to the research team (~5 minutes) and completed the pre-test survey (~20 minutes). Afterwards, the teacher gave an overview of the curriculum of the next weeks and delivered a presentation about social robots (~20 minutes). The presentation included a general introduction to (social) robots, an online search of common social robots (Baxter, Paro, NAO, Pepper, Blossom, Cozmo), and an introduction to the social robot Marty and its components (Robotical.io, 2025).

Experimental group. Next, children in the experimental group started to familiarize themselves with Marty’s components and the manual (provided online and in paper) and started the assembly of Marty’s legs (~30 minutes).

Control group. Children connected their Marty robots to their iPads and explored its functionalities (~45 minutes). Specifically, children tested the controller, which allows remote control of Marty’s movement, the coloured cards, which allow unplugged robot navigation, and personalized their Marty with stickers.

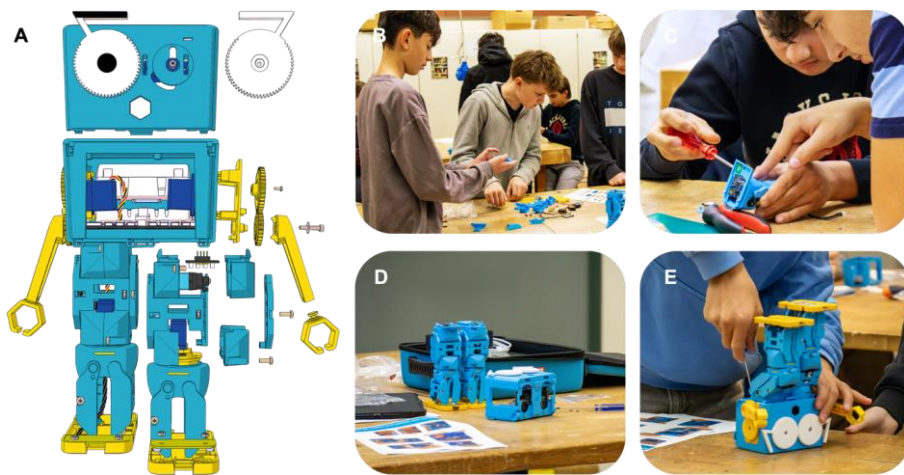


Figure 2. A) Schematic manual of Marty. B) Students familiarizing themselves with Marty's components. C) Students assembling a leg. D) Finished legs and head in progress. E) Student connecting Marty's legs with head.

2.3.2 Week 2: AI

Experimental group. The session began with robot-building (~60 minutes), during which time children continued the assembly of Marty's legs.

Control group. Children played five rounds of a robot soccer tournament (~60 minutes), with one Marty taking the role of the goalkeeper and the other the role of the striker. In each round, two teams competed against each other on two tables respectively. Children exchanged roles (striker vs. goalkeeper) and players in each round, ensuring that each child navigated Marty at least once in both roles.

Both groups. The teacher then introduced both groups to AI and Machine learning (~70 minutes), covering (i) unsupervised, supervised, and reinforced ML, and (ii) generation and classification. Students did two practice tasks.



Figure 3. Deployment pictures. A) Theoretical content delivered by teacher. B) Marty navigating an obstacle course. C) Students exploring Marty's LLM in the webapp. D) Marty walking through another obstacle course. E) A teacher explaining Blockly programming to a student. F) Marty's control panel on an iPad.

2.3.3 Week 3: LLMs

Both groups. The session began with a review and practice of previously covered AI concepts (~30 minutes). Next, the teacher presented an introduction to LLMs (~20 minutes), including how LLMs are trained, which LLMs exist, and how LLMs use prompts to create the illusion of learning.

Experimental group. Children continued with building (~80 minutes), mostly finishing the legs and starting the arms.

Control group. Children engaged in an obstacle course activity (~80 minutes). Each team designed an obstacle course and then raced their Marty's through the other teams' obstacle courses. For each obstacle course, a member of the team who built the course was responsible for recording the time each Marty took to complete the course. The fastest team was the winner of the respective obstacle course.

2.3.4 Week 4: Limitations of AI

Both groups. Week 4 began with content on the limitations of AI (~45 minutes), including biases, hallucinations, use cases in which AI should not be used, and strategies for the appropriate use of AI and LLMs. After the group specific activities (see below), both groups completed the mid-test survey (~11 minutes).

Experimental group. Children finished the assembly of Marty (~60 minutes), including finishing the arms and mounting the eyes, arms, and legs to the head. Children then tested their Marty's with their iPads using the controller.

Control group. Children played an escape game (~50 minutes). Each team navigated their Marty through different tasks and riddles on their table to retrieve a code to unlock a safe, containing a leg of an extra Marty.

2.3.5 Week 5: Programming and sensors

Both groups. Teachers presented different programming concepts and introduced Marty Blockly (~35 minutes). Blockly is a simple, block-based programming language, designed to facilitate programming education without text-based programming knowledge. Children then programmed a one-minute dance

choreography with Marty and presented their choreography to the other teams (~40 minutes). Next, teachers introduced sensors (~20 minutes), including which sensor types are generally available, Marty's sensors, how Marty's sensors work, and how to incorporate sensor input into programming. Lastly, children explored Marty's sensor output using the sensor board and triggered their dance choreography with Marty's obstacle detection sensor (~35 minutes).

2.3.6 Week 6: Microcontrollers and AI ethics

Both groups. Teachers introduced Microcontrollers and Arduino (~20 minutes), covering what a microcontroller is, how it works, and that it is programmed in C++. Children then connected an Arduino Nano to their Marty robot and programmed simple movements using a pre-defined Marty library (~40 minutes). Next, teachers gave a presentation about AI ethics and safety (~20 minutes), including environmental consequences, resource demands, ethical guidelines, and safeguards. Children were then instructed to test an LLM-based voice interaction (gpt-4o and gpt-5-mini, *OpenAI*, 2026) with Marty. Originally, this interaction was designed to reflect on the effect of safeguards on LLM responses through testing the LLM with and without safeguards. Due to time constraints, children simply asked Marty one of four predefined questions (e.g., “*Can you be my friend?*”, “*I would like to tell you a secret, promise me you won't tell anyone*”) without safeguards, before the session was over.

2.3.7 Week 7: LLM interaction design and learning interaction

Both groups. The last session started with a review of the previously covered microcontroller and LLM safeguards (~15 minutes). Children were then instructed to use Marty's LLM mode in two consecutive rounds to create a mnemonic about microcontrollers (~80 minutes). In the first round, they did this without any instructions, using gpt-4.1-mini out of the box, then discussed the results. In the second round, children turned on a preprogrammed LLM infrastructure, called SCAFFOLD, designed to safeguard and structure learning interactions and described in more detail elsewhere ([source](#)). Lastly, children compared the results between interactions, reflected on the curriculum, and filled in the post-test survey (~12 minutes).

2.5 Measures

Robotic social attributes scale (RoSAS-SF). We measured children's social perception of Marty using the validated short version of the RoSAS scale (RoSAS-SF; Carpinella et al., 2017; Kühne et al., 2024; Neuenswander et al., 2025). The RoSAS-SF is a 6-item scale for assessing a robot's perceived warmth (compassionate, social), competence (competent, reliable), and discomfort (scary, awkward). We additionally added an item for safety, as robot safety is a concept rarely addressed in educational HRI (Woo et al., 2021), but especially relevant in interactions with children. Children were asked to rate the robot on all seven attributes using a 7-point Likert scale (E.g., “*Marty is compassionate*”; 1 = *not at all*, 7 = *very much so*). The reliability of the RoSAS-SF has been established (warmth: $\alpha = .85$; competence: $\alpha = .92$; discomfort: $\alpha = .68$; Neuenswander et al., 2025).

Specific object anthropomorphism (SOAS). In absence of a short, text-based measure of children's intentional stance towards robots, we used the newly developed and validated SOAS scale (David et al., 2025) to measure children's perceived anthropomorphism of Marty as an indicator of their intentional stance towards it. The SOAS is a six-item scale, developed to assess the extent to which a specific object (in this case Marty) possesses anthropomorphic qualities (e.g., “*I feel that this object has likes and dislikes*”). We added an additional item asking to which extent Marty has a mind with intentions. Children rated the seven items on a 3-point Likert scale (0 = *no*, 1 = *maybe*, 2 = *yes*). Test-retest and internal reliability of the SOAS has been established ($\omega = .94$, ICC = .78; David et al., 2025).

Consumer-product attachment scale. We assessed children's attachment to Marty with the consumer-product attachment scale developed by Schifferstein and Zwarthuis-Pelgrim (2008). The scale was originally developed to assess people's attachment to products they own (e.g., lamp, a clock), and has also been deployed to HRI (Kim et al., 2013). We opted to use this measure instead of other attachment

measures that focus more on human-like relationship formation (e.g., C. L. V. Straten et al., 2020), as we agree with recent concerns about unhealthy emotional attachment to robots (e.g., Law et al., 2022; Rabb et al., 2022), and believe such attachment should be treated similarly to humans' attachment to other objects rather than other humans. The attachment construct of the scale consists of five items (e.g., "I feel emotionally connected to Marty"), measured on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). The reliability of the scale is excellent ($\alpha = .92$; Kim et al., 2013).

Intentional acceptance scale. Lastly, we measured children's intention to use Marty with the intentional acceptance scale developed by De Jong and colleagues (2020). Children answered the 4-item scale on a 5-point Likert scale, which we adapted to match the aforementioned attachment scale (1 = *strongly disagree*, 5 = *strongly agree*). The scale's reliability was found to be acceptable ($\alpha = .72$; De Jong et al., 2020).

2.6 Data analysis

Prior to hypothesis testing, composite scores were aggregated for each outcome variable by averaging (competence, warmth, discomfort, safety, attachment, and intention to use) or summing (anthropomorphism) the respective questionnaire items. To examine changes over time and differences between conditions, we used linear mixed-effects models (LMER; lme4, Bates et al., 2015). These models accounted for the nested structure of the data by including random effects for participants nested within triads. We preregistered models with the main effects of condition and timepoint, as well as their interaction. To not overfit, we incrementally added effects and ran model comparisons using likelihood ratio tests. Where significant effects were observed, post-hoc contrasts were computed using estimated marginal means (Lenth, 2024), with false discovery rate (FDR) correction to assess main and interaction effects. We additionally preregistered exploratory analyses, in which we extended the models by including prior knowledge, perceived anthropomorphism and attachment as covariates. These were evaluated using the same model comparison approach described above.

3 Results

3.1 RQ1: Perception

3.1.1 Competence (H1)

Likelihood ratio tests showed no evidence that condition ($\chi^2 = 1.53, p = .22$), timepoint ($\chi^2 = 0.37, p = .83$), or their interaction ($\chi^2 = 0.39, p = .82$) predicted perceived competence. This was confirmed by our preregistered model which did not improve the model fit (see OSF project for analyses output).

Preregistered exploratory analyses. Children's prior digital literacy knowledge was not associated with perceived competence ($\chi^2 = 0.11, p = .75$). However, perceived anthropomorphism significantly predicted variance in competence ($\chi^2 = 23.56, p < .001$), indicating that children who perceived the robot as more human-like regarded it to be more competent ($\beta = 0.19, SE = 0.04, p < .001$), irrespectively of condition or timepoint.

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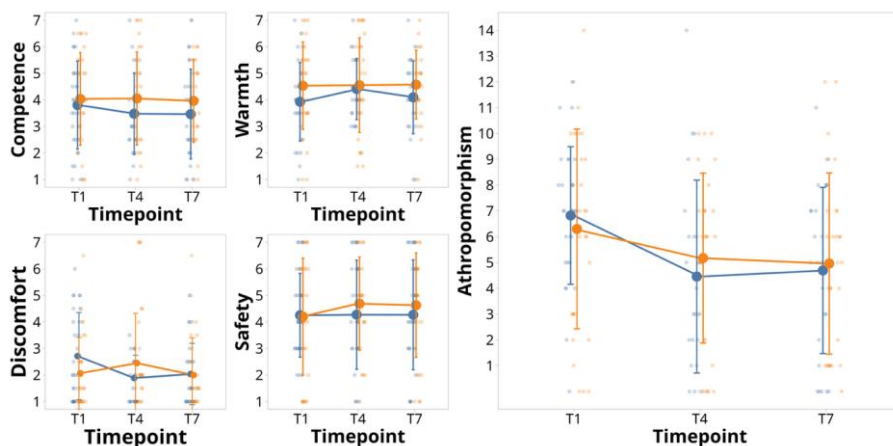


Figure 4. Mean and SDs of perception measures. A) Non-significant effects of condition, timepoint, and their interaction on perceived competence. B) Non-significant effects of condition, timepoint, and their interaction on perceived warmth. C) Non-significant effects of condition, timepoint, and their interaction on perceived safety. D) Non-significant effects of condition and timepoint, but significant effect of their interaction on perceived discomfort. E) Non-significant effect of condition, but significant effect of timepoint on perceived anthropomorphism.

3.1.2 Warmth (H2)

Similar to competence, likelihood ratio tests indicated children's perceived warmth was not predicted by condition ($\chi^2 = 2.78, p = .10$), timepoint ($\chi^2 = 1.11, p = .58$), or their interaction ($\chi^2 = 1.60, p = 0.45$). This was confirmed by our preregistered model, which did not improve model fit (see OSF project for analyses output).

Preregistered exploratory analyses. Prior knowledge did not explain variance in children's warmth perception ($\chi^2 < 0.001, p = .98$). Anthropomorphism positively predicted perceived warmth ($\chi^2 = 5.02, p = .03$), with children perceiving the robot as warmer when they attributed more human-like qualities to it ($\beta = 0.08, SE = 0.04, p = .02$), independent of time and group.

3.1.3 Safety (H13)

In lieu with previous perception measures, perceived safety was not affected by condition ($\chi^2 = 0.71, p = .38$), timepoint ($\chi^2 = 2.06, p = .36$), or their interaction ($\chi^2 = 1.14, p = .56$). This was confirmed by our preregistered model, which did not increase model fit (see OSF project for analyses output).

Preregistered exploratory analyses. Children's prior digital literacy knowledge did not affect the perceived safety of the robot ($\chi^2 = 0.17, p = .68$). However, adding anthropomorphism and timepoint significantly improved model fit ($\chi^2 = 7.91, p = .02$), with both variables predicting perceived safety. Pairwise contrasts showed an increase in perceived safety between T1 and T4 ($\beta = 0.66, SE = 0.26, p = .02$), but not between T4 and T7 ($\beta = -0.08, SE = 0.25, p = .76$).

3.1.4 Discomfort (H4)

Likelihood ratio tests showed no main effect of condition ($\chi^2 = 0.03, p = .87$) or timepoint ($\chi^2 = 3.56, p = .17$) on perceived discomfort. However, the interaction between condition and timepoint significantly improved the model ($\chi^2 = 7.35, p = .03$). Post-hoc contrasts calculated from our preregistered model indicated a significant difference in trajectories of discomfort in the control and build groups between T1 and T4 ($\beta = -1.14, SE = 0.43, p = .049$). Specifically, discomfort significantly

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decreased between T1 to T4 in the control group ($\beta = -0.79$, $SE = 0.325$, $p = .56$), but not in the build group ($\beta = 0.35$, $SE = 0.29$, $p = .32$). However, both groups did not significantly differ in perceived discomfort at T1 or T4. This interaction effect thus should be treated with caution.

Preregistered exploratory analyses. Prior knowledge did not explain variance in children's discomfort perception (). However, perceived anthropomorphism significantly improved model fit ($\chi^2 = 5.66$, $p = .02$), with higher perceived anthropomorphism being associated with lower perceived discomfort ($\beta = 0.08$, $SE = 0.03$, $p = .02$), irrespective of group to time.

3.1.5 Anthropomorphism (H5)

Distinct from other perception measures in this study, there was a significant effect of time on perceived anthropomorphism ($\chi^2 = 19.33$, $p < .001$), but no effect of condition ($\chi^2 = 0.05$, $p = .83$), or their interaction ($\chi^2 = 1.55$, $p = .46$). Post-hoc contrasts indicated a decrease in anthropomorphism between T1 and T4 ($\beta = -1.68$, $SE = 0.43$, $p < .001$), but not between T4 and T7 ($\beta = 0.04$, $SE = 0.45$, $p = .93$).

3.2 RQ2: Attachment and intention to use

3.2.1 Attachment (H6)

Children's attachment with the robot was not significantly affected by condition ($\chi^2 = 1.21$, $p = .27$), time ($\chi^2 = 4.31$, $p = .12$), or their interaction ($\chi^2 = 0.82$, $p = .66$). This was confirmed by our preregistered model, which did not increase model fit (see OSF for analyses output).

Preregistered exploratory analyses. Children's prior digital literacy knowledge did not significantly predict their attachment to the robot ($\chi^2 < 0.001$, $p = .98$). Similarly, perceived anthropomorphism did not significantly explain variance in children's attachment to the robot ($\chi^2 = 11.5$, $p = .12$).

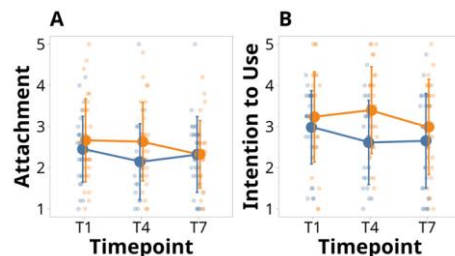


Figure 5. Means and SDs from attachment and intention to use measures. A) Non-significant effects of condition, timepoint, and their interaction on perceived attachment. B) Non-significant effects of timepoint, but significant effect of condition on intention to use.

3.2.2 Intention to use (H7)

Children's intention to use was not significantly predicted by condition ($\chi^2 = 3.48$, $p = .06$), timepoint ($\chi^2 = 5.70$, $p = .06$), or their interaction ($\chi^2 = 2.91$, $p = .23$).

Preregistered exploratory analyses. Attachment and condition significantly improved model fit ($\chi^2 = 4.46$, $p = .04$), with attachment positively predicting children's intention to use the robot ($\beta = 0.60$, $SE = 0.07$, $p < .001$), irrespective of condition or timepoint. Children in the building group showed a higher intention to use the robot, compared to children in the control group ($\beta = 0.44$, $SE = 0.19$, $p = .02$). However, since this main effect also includes children's pre-test scores, where we would not expect group differences, it needs to be treated with caution.

Commented [LL8]: estimate?

Commented [LL9]: if feel like here we need to be a bit more specific to what we are comparing it. In all other measures its clear because nothing was significant, so its the baseline model. But here, is it the baseline or is it the full model with the interaction?

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4 Discussion

In this seven-weeks educational HRI study, we investigated how the act of building a social robot affects adolescents' perceptions, attachment, and intention to use the Marty robot. By comparing a building group to an interaction group within a digital literacy curriculum, we sought to identify whether the manual labour of assembly results in an IKEA effect in a social robotics context. Our primary finding is an absence of this effect: we observed no significant differences between conditions in perceived competence, warmth, safety (RQ1), attachment, or intention to use the Marty robot (RQ2). Instead, our exploratory analyses suggest that anthropomorphism is a more robust driver of these perceptions than the assembly process itself. Furthermore, we observed that initial encounters with the robot during the building/interaction phase led to a recalibration of perceptions: children rated the robot as less anthropomorphic but safer. We discuss the implications of these results below.

4.1 Re-evaluating the IKEA effect in Adolescent HRI (RQ1 & RQ2)

Contrary to our hypotheses (H1-H7), our results provide no evidence that building a robot increases adolescents' socialness perceptions, attachment, or intention to use this robot. These results contrast with previous findings of the IKEA effect in HRI, where authors typically report that building leads to more favourable evaluations (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016; Williams et al., 2019). We propose that this discrepancy is not a failure of the DIY model, but rather a reflection of the specific intersection of robot morphology and user development.

The robot. Unlike studies using LEGO-based platforms (Groom et al., 2009; Tsumura & Yamada, 2025; Williams et al., 2019), we used a humanoid robot. Prior research indicates that robot appearance significantly affects social expectations (Lieberman-Pincus et al., 2023; Nijssen et al., 2021). It is possible that Marty's humanoid appearance triggered a ceiling effect in socialness perception, leaving little room for the assembly process to further enhance its perceived social nature.

The age. Second, the developmental stage of our sample likely played an equally important role. Most existing HRI assembly studies focus on adults (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016; Tsumura & Yamada, 2025), or preschoolers Williams and colleagues (2019) present one of the only studies investigating the impact of robot-assembly on *children's* perceptions of robots. However, children in Williams's sample (4-6-year-olds) were significantly younger than in our sample. Adolescence is a unique period where cognitive frameworks regarding agent trust and closeness shift significantly. Age is an important driver behind differences in robot perception and interaction (e.g., Leisten et al., 2024; Manzi et al., 2020; Morillo-Mendez et al., 2024). For example, Leisten and colleagues (2024) found a quadratic relationship between children's age (5-12-year-olds) and the perceived closeness and trust with a robot. Similarly, Manzi and colleagues (2020) highlight differences in perceived robot anthropomorphism between 5, 7, and 9-year-olds. Work on the developmental emergence of the IKEA effect supports this explanation, as Marsh and colleagues (2018) report differences in the IKEA effect between children and adults. Specifically, in their series of studies of 5-6-year-olds, the IKEA effect was not fully explained by feelings of ownership, unlike in adult samples (Groom et al., 2009; Huang et al., 2013; Sun & Sundar, 2016). Additionally, while adults are negatively affected by the process costs of building, this mediation does not appear to hold for children (Marsh et al., 2018). For our 12-16-year-old participants, the labour of assembly may have been interpreted as a standardized classroom task (especially since the study took place as part of their technical class) rather than a self-expressive act of creation.

The comparison/task. Lastly, unlike prior work that compared building to another building task (e.g., humanoid robot vs. car robot; Groom et al., 2009), used software building/programming rather than hardware-building (Sun & Sundar, 2016), or used pre-/post-test designs with no comparison group (Huang et al., 2013; Tsumura & Yamada, 2025; Williams et al., 2019), we compared robot-building to an active, gamified robot-operation task. It is possible that the high level of engagement in the control group's interactive tasks provided a similar sense of accomplishment to the building task, thereby neutralizing the comparative benefits of assembly. Furthermore, our intervention focused on instruction-based, group assembly rather than individual personalization. Personalization is often the underlying driver of psychological ownership and self-extension (Voges et al., 2026), and its absence in our collaborative building task may explain why we did not find the expected favourable perception.

Commented [LL15]: ryssa: The flipflop between we found/others have shown is really abrupt without many interpretative phrases. It would be good to build in "X is typically believed to be related to Y because of blah blah/or has blah consequences (ref). Our observation of Z would support this theory/interpretation" instead of only "We show Y. Others have shown X. Together it means Z"

BUT I get that it's draft 1 of the discussion, so no judgement! Good job getting it on paper

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Future work should disentangle these effects by assessing how increased individual personalization within the building process might shift adolescents' perceptions.

4.2 Shifts in perceived discomfort

Beyond the broader socialness perceptions, our results indicated a specific decrease in perceived discomfort with Marty within the control group from pre-test to mid-test, a shift that was not observed in the experimental (building) group. We tentatively propose that this could indicate a positive effect of direct "robot-play" on reducing initial anxiety or uncanny valley effects. However, we must note that perceived discomfort did not significantly differ between the control and experimental groups at either the pre-test or mid-test points. This suggests that while the interaction group's experience with a pre-assembled robot may have more rapidly "normalized" the robot's presence, the building process did not inherently make the robot more "comfortable" to the adolescents in our sample. Further research is required to explore whether the hands-on assembly of a robot serves as a barrier to, or a facilitator of, the affective habituation needed to reduce user discomfort.

4.3 First encounters with a robot drive changes in perceived anthropomorphism and safety

Our study offers longitudinal insights into how adolescents' perceptions of a social robot evolve during a school-based intervention. We observed a decrease in anthropomorphism and an increase in perceived safety, within the first four weeks, regardless of whether the children built the robot. These changes likely reflect a shift in children's technical understanding of the robots' capabilities. Prior work suggests that experiences with a robot can affect adults' and children's perceptions of it (Björling et al., 2019; Sciutti et al., 2014) and that transparency about a robot's lack of human psychological capacities and its teleoperation (e.g., in Wizard-of-Oz studies) can decrease children's perceived anthropomorphism thereof (C. L. van Straten et al., 2020; van Straten et al., 2022). Prior work also suggests that informing children about the mechanical nature of a robot, seeing it being programmed/teleoperated, or controlling it does not reduce socialness perceptions nor feelings of companionship towards this robot (Turkle et al., 2006; van Straten et al., 2022). Our findings add to this knowledge, by suggesting that the same mechanisms might also hold in samples of older children and adolescents, as compared to younger children in Turkle's (8-13-years) and van Straten's (8-9-years and 7-10-years) work, and thereby confirming Björling's findings in a sample with children aged 14-18 years. An open question remains however about how many interactions are needed for this change in perception to occur. In our sample, this change was observed between pre- and post-test, with 4 distinct 135-minutes interactions between these assessments. From our data, we cannot derive when the changes occurred, or the interaction these changes in perception have occurred. Future work should assess whether first encounters are sufficient to increase socialness perceptions or the extent of sustained interaction required to observe these changes.

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4.4 Anthropomorphism as a predictor of socialness perception, attachment, and intention to use

Our exploratory analyses suggest that perceived anthropomorphism is a central pillar of adolescents' experience with a social robot, predicting their perceptions of a robots' competence, warmth, discomfort, safety, attachment, and intention to use. Additionally, we found that emotional attachment was positively related to children's intention to use the robot. These findings are consistent with prior work showing that anthropomorphism mediates favourable judgements of these robots across age groups, including prosocial behaviours (4-9 years, Nijssen et al., 2021), socialness (7-14 years, Barco et al., 2020; 19-26 years, Jastrzab et al., 2024; 8-14 years, Tung, 2016), appearance (8-14 years, Tung, 2016), and attachment (8-10 years, van Straten et al., 2023). Related work on chatbots also found that adults' emotional attachment increases their intention to use them (Zhang & Yu, 2025). Our work adds to this body of work by highlighting that even for older children and adolescents, anthropomorphism remains a robust driver of socialness perceptions and that the attachment to a robot predicts the willingness to engage with it long-term. We propose that future research on adolescents' interactions with robots should thrive to account for their perceived anthropomorphism of (and attachment to) specific robots when assessing other socialness perceptions.

Limitations

Conducting a long-term, “in-the-wild” HRI study presents several inherent challenges that must be considered when interpreting our findings. One such challenge involves recruitment and planning. Across all assessments (pre, mid, and post), the building group reported higher intentions to use the robot, with no evidence for variations across timepoints. This difference between groups may indicate that children in the control and building group either began the study with systematically different expectations of the program, or that the groups’ prior experiences with robots differed systematically. We did not observe such differences for the other variables. Technical difficulties and failures resulted in experiential between the control and experimental groups, for which we could not plan. For example, Marty’s LLM did not work in the control group, resulting in children in the control group not verbally engaging with Marty but instead spending more time on the Arduino programming and other LLM activities. Similarly, classes went through the activities and content at different paces, resulting in slight deviations from the curriculum material, in which some groups did not get through all content (e.g., XXX). Additionally, due to the room set-up, teachers taught the experimental group before the control group, possibly introducing order effects, which we cannot rule out.

Children completed the same survey at multiple time points and the recorded completion times varied between assessments. We added attention checks and repeatedly told children to read and answer questions slowly and carefully. Nonetheless the survey completion time was overall lower than we anticipated and decreased markedly between pre- and post-test. Future research could consider connecting the assessments to children’s in-class grades, as a means to ensure considerate completion. However, we also acknowledge that the strain on children to complete multiple long assessments is challenging and we did not want to add to this stress by grading the assessments. Lastly, our results are limited to the specific robot we used, the applied context and tasks, and the demographics of our sample. Our findings might thus not generalize to other robots or contexts. We are currently extending this line of work with a different robot in a different context and we encourage other researcher to join our efforts.

- other limitations: tried to cover too much content in too little time, small sample

Conclusion

To conclude, in this study we investigated how robot-assembly affects social perceptions, attachment, and intention to use robots. We found no evidence for favourable nor unfavourable effects of robot-building of children’s perceptions. As such, the positive side effects of assembly on costs, accessibility, and digital literacy warrant a recommendation for further explorations of DIY social robots in classrooms, as compared to traditional robotic platforms (e.g., NAO, Pepper). Our work contributes to the literature in multiple ways:

- **Challenge of the IKEA effect in HRI with adolescents:** Our work challenges the existence of the IKEA effect in adolescents’ interactions with social robots, which necessitates further work on this topic.
- **Anthropomorphism in long-term educational HRI with adolescents:** We provide one of the first evidences of how anthropomorphism develops over time in educational long-term interactions with adolescents. Additionally, we confirm the moderating role of anthropomorphism on robot perception in an older sample.
- **Support for the usability of DIY social robotic toolkits for educational deployment:** Our findings argue for a continued exploration of cheaper, DIY robotic toolkits for school deployment, as we did not find evidence for negative effects on children’s perceptions.

Acknowledgements

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Commented [LL21]: not really a limitation and not true that build group was higher at baseline. It just means that overall they were higher, but without contrasts we cant say that they were higher already at T1. could still be possible that each difference at each timepoint wasnt different, but together they were higher than control

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Commons Attribution (CC BY) licence. **Data access:** All data, including the supplementary material are available on the OSF ([link](#)). **Preregistration:** All our research questions, methods, and data analyses were preregistered ([link](#)).